SERVO BARRIER SYSTEM USING ARDUINO

ABSTRACT

The Servo Barrier System is an automation project designed to open or close a barrier based on distance measurement using an ultrasonic sensor. When an object or person approaches within a specified range, the servo motor automatically lifts the barrier, and when the object moves away, the barrier closes again. The system is controlled by an Arduino UNO microcontroller, making it an excellent example of smart access automation. This project is commonly used in automated gates, parking systems, toll booths, and security entrances.

OBJECTIVE

The objective of this project is to design and develop an automatic barrier control system that uses an ultrasonic sensor to detect approaching objects and operates a servo motor without manual effort, increasing security, convenience, and automation efficiency.

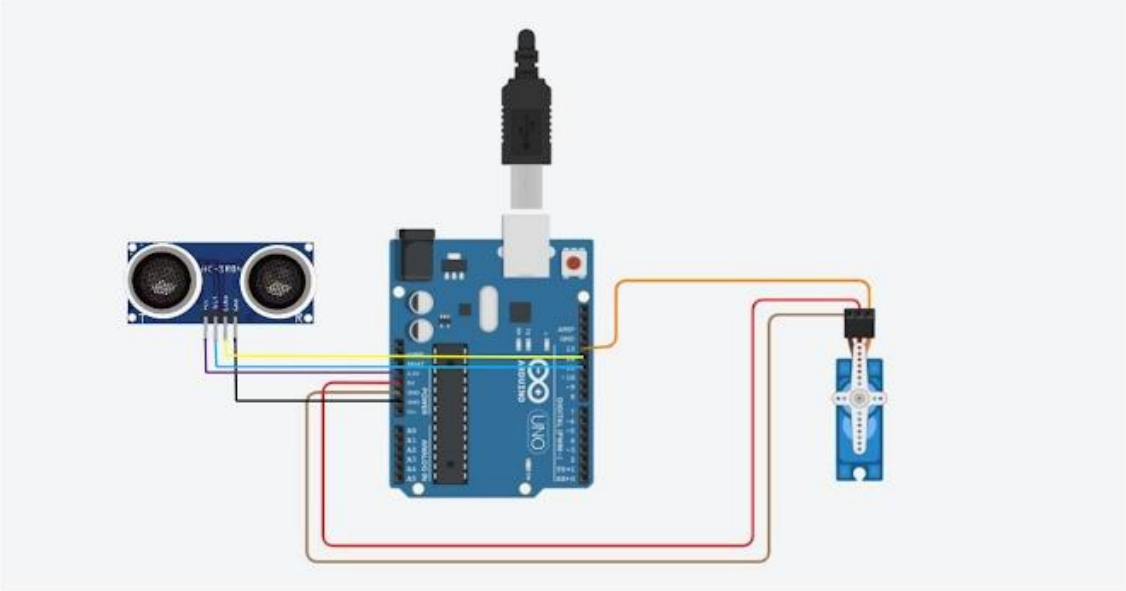
COMPONENTS REQUIRED

S.No	Component	Quantity
1	Arduino UNO	1
2	Ultrasonic Sensor (HC-SR04)	1
3	Servo Motor (SG90)	1
4	Breadboard	1
5	Jumper Wires	As required
6	USB Cable	1
7	Power Supply (5V from Arduino)	1

CIRCUIT DESCRIPTION

The ultrasonic sensor measures the distance of an incoming object by sending ultrasonic pulses and receiving the reflected waves. The sensor outputs a calculated distance to the Arduino, which then compares the value with a predefined threshold. If the detected distance is less than 20 cm (indicating an object close to the barrier), the Arduino triggers the servo motor to raise the barrier. When the distance increases, the servo motor rotates back to close the barrier.

CIRCUIT DIAGRAM



CIRCUIT CONNECTIONS

Component	Arduino Pin
Ultrasonic Sensor VCC	5V
Ultrasonic Sensor GND	GND
Ultrasonic Sensor TRIG	D9
Ultrasonic Sensor ECHO	D10
Servo Motor VCC	5V
Servo Motor GND	GND

Component	Arduino Pin
Servo Motor Signal	D11

ARDUINO CODE

```
#include <Servo.h>

#define TRIG_PIN 9

#define ECHO_PIN 10

#define SERVO_PIN 11

Servo myServo;

void setup() {

pinMode(TRIG_PIN, OUTPUT);

pinMode(ECHO_PIN, INPUT);

myServo.attach(SERVO_PIN);

Serial.begin(9600);

}

long getDistance() {

digitalWrite(TRIG_PIN, LOW);

delayMicroseconds(2);

digitalWrite(TRIG_PIN, HIGH);

delayMicroseconds(10);

digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH);

  long distance = duration * 0.034 / 2; // distance in cm

  return distance;

}

void loop() {

  long distance = getDistance();

Serial.print("Distance: ");
```

```
Serial.print(distance);

Serial.println(" cm");

  // Example action: move servo based on distance

  if (distance < 20) {

myServo.write(180);

  } else {

myServo.write(0);

  }

delay(500);

}
```

WORKING PRINCIPLE

The system works on the principle of **ultrasonic distance measurement**. The ultrasonic sensor continuously emits sound pulses and calculates the time taken for the waves to return after hitting an object. Based on this, it determines the distance. When an object is nearby (less than the threshold distance), the servo rotates to lift the barrier. When the object leaves, the servo returns to its initial position.

RESULT & OUTPUT

Once the circuit is powered ON and the program is successfully uploaded to the Arduino, the automatic barrier system begins functioning immediately. When the ultrasonic sensor detects an object or person approaching within a distance of 20 cm, the servo motor activates and the barrier arm lifts automatically to allow passage.

After the object moves away and the distance increases beyond the threshold, the system detects that the path is clear and the barrier arm returns smoothly to its closed position.

Throughout the operation, the Serial Monitor displays continuous real-time distance readings from the ultrasonic sensor, helping to observe the detection accuracy and system response. The output confirms that the project works reliably and consistently under normal operating conditions.

APPLICATIONS

• **Parking Gates** – The system can be used to automatically open and close barriers at parking lot entrances, improving vehicle flow and reducing the need for manual operation.

• **Toll Plazas** – Automated barriers controlled by sensors help manage traffic efficiently at toll booths, ensuring quick entry and exit for vehicles.

• **Automated Security Entrances** – Useful in secured areas such as offices, apartments, and industrial sites where controlled access is required without human supervision.

• **Railway Crossing Models** – Ideal for demonstrating automatic gate operation in railway projects or miniature railway setups to enhance safety.

• **Smart Home and Building Automation** – Can be integrated into smart systems for controlling gates, garage doors, and restricted areas based on proximity detection

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ADVANTAGES

• **Reduces Manual Effort** – The automatic barrier system operates without the need for human intervention, making access control more convenient and efficient.

• **Enhances Automation and Security** – By detecting objects accurately and controlling the barrier automatically, the system increases safety and reduces the chances of unauthorized entry.

• **Low-Cost and Energy Efficient** – Built using affordable components like Arduino and ultrasonic sensors, the system consumes minimal power while delivering effective performance.

• **User-Friendly and Reliable** – The setup is simple to install, operate, and maintain, ensuring consistent functioning in various environments.

• **Expandable with IoT or RFID** – The system can be easily upgraded by adding technologies such as IoT modules, RFID readers, or wireless control to enable remote monitoring and advanced access management.

CONCLUSION

The Automatic Servo Barrier System successfully demonstrates real-time object detection and barrier control using an ultrasonic sensor and Arduino. The project highlights the practical implementation of sensors and microcontrollers in automation and serves as an excellent base for industrial and smart city applications.